CRYSTAL VMM SERIES COMPARISON

Specification	Manual VMM	Semi Automatic VMM	Fully Automatic VMM	Automatic VMM with Cabinet	VMS VMM
Model Series	AA-9011M / AA-9012M / AA-9013M	AA-9021SA / AA-9022SA / AA-9023SA	AA-9031A / AA-9032A / AA-9033A	AA-9041AC / AA-9042AC / AA-9043AC / AA-9044AC	AA-9051V / AA-9052V / AA-9053V
Dimension (mm)	540×540×860 / 665×670×915 / 724×855×915	550×540×930 / 600×740×980 / 700×840×980	540×830×1080 / 640×830×1080 / 740×980×1080	650×950×1740 / 730×1040×1740 / 790×1130×1760 / 890×1370×1780	550×540×880 / 600×740×930 / 700×840×930
Measuring Range (mm)	200×100×150 / 300×200×200 / 400×300×200	200×100×150 / 300×200×200 / 400×300×200	200×200×200 / 300×200×200 / 400×300×200	300×200×200 / 400×300×200 / 500×400×200 / 600×500×200	200×100×150 / 300×200×200 / 400×300×200
Measuring Accuracy (µm)	2.5 + L/100	2.5 + L/100 , 2.6 + L/100	2.5 + L/100	2 + L/100 , 2.5 + L/100	2.5 + L/100
Repeatability (µm)	2.5	±2.5 / ±3	2.5	2 / 2.5	2.5
+Weight (kg)	150 / 200 / 255	130 / 180 / 230	140 / 180 / 230	380 / 530 / 640 / 715	140 / 190 / 240
CCD	Industrial-grade color CCD camera	Industrial-grade color CCD camera	1,300,000 pixels industrial-grade color CCD camera	1.3MP digital camera	Industrial grade CCD camera
Lens	Manual zoom objective 0.7–4.5X	Manual lens 0.7–4.5X	Automatic zoom lens	1–10X automatic zoom lens	Manual dented zoom lens 0.7–4.5X
Magnification	18–195X	18–195X	24.7–160X	18–195X	18–188X
Field of View (mm)	8.1–1.3	8.1–1.5	8.5–1.3	8.1–1.3	8.1–1.3

Working Distance (mm)	80	80	80	80	108
Linear Scale Resolution	0.5 µm (0.1 µm optional)	0.5 µm (0.1 µm optional)	0.5 µm (0.1 µm optional)	0.5 (0.1 optional)	0.001 mm
Driving System	X/Y/Z axis manual control	Manual X/Y, CNC Z-axis control	CNC control with high-precision servo motors	CNC control with high-precision servo motor drive	X/Y/Z axis manual control
Illumination	Programmable LED 8-section ring surface light + LED bottom parallel light	Programmable LED 8-zone ring surface light + LED bottom parallel light	Programmable LED 8-zone ring surface light + LED bottom parallel light	Programmable LED 8-zone ring surface light + LED bottom parallel light	LED surface light + parallel LED bottom light



Manufactured by

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